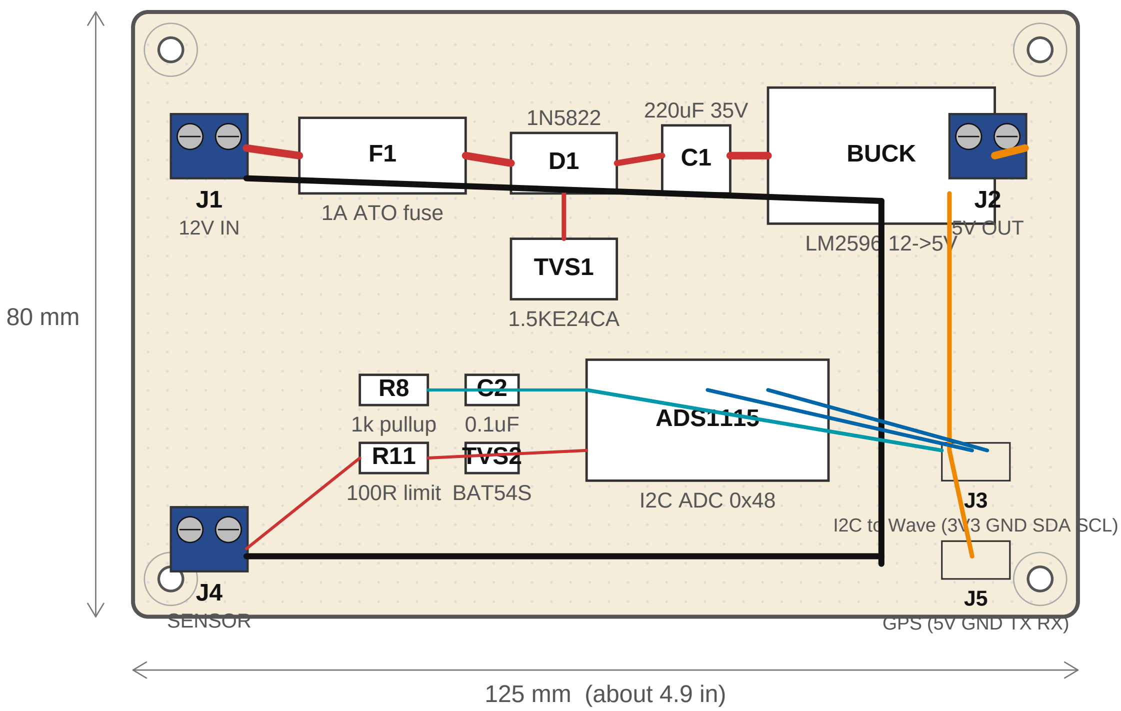
Page 1 of 4 -- Single-Board Hand-Build (Perfboard)

125 x 80 mm prototype perfboard. 2.54 mm hole grid shown. Component placement is the assembly map -- traces shown are wires

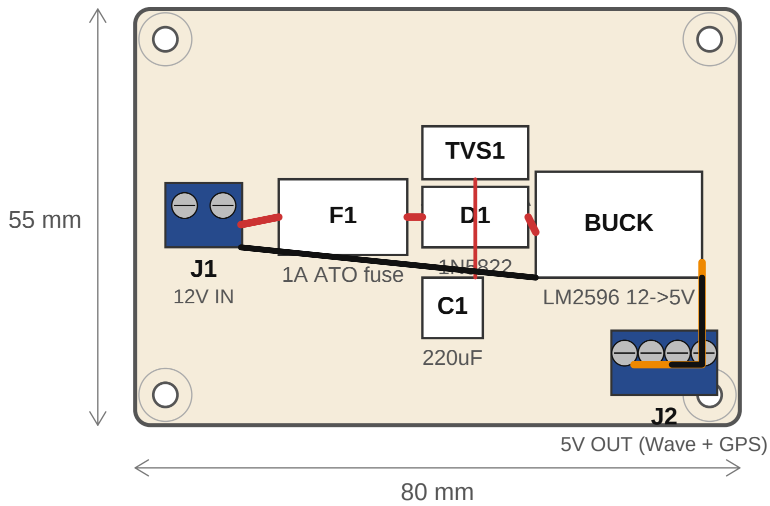


Trace conventions:

- **+12V** 1.0 mm trace -- 1A continuous
- **+5V** 1.0 mm trace -- 1A continuous
- **+3.3V** 0.5 mm trace -- low current to ADC
- **I2C** 0.5 mm trace -- digital signal
- **Sense** 0.4 mm trace -- low-current analog
- **GND** 0.8 mm; star ground at one common point

Page 2 of 4 -- Split Design / Board A: Power Board

80 x 55 mm. EasyEDA-friendly layout. 12V input -> F1, TVS1, D1, C1 -> BUCK -> 5V output (to Waveshare and GPS).



BOARD A NOTES:

* J2 is a 4-position 3.5 mm screw terminal -- two paralleled +5V/GND rails.
  Wire one to Waveshare, the other to the GPS module.

* C1 placed CLOSE to the BUCK input pin so it can absorb cranking-dip transients
  before they reach the regulator.

* TVS1 sits in shunt -- one lead to +12V rail, one to GND, with shortest possible
  path. Place BEFORE D1 so it can clamp the unprotected input directly.

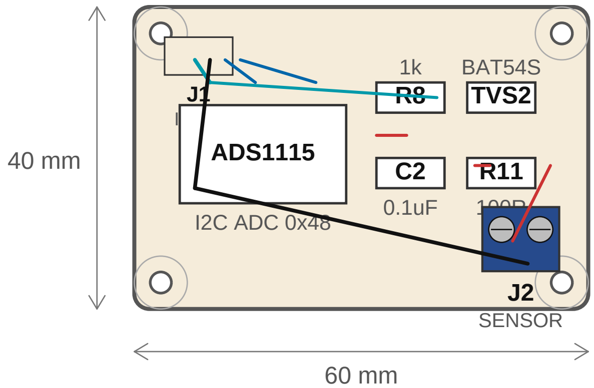
* Use a 1.5 mm or wider trace from F1 to BUCK input (the high-current path).

PROTECT-FIRST RAIL ORDER:

+12V IN -> F1 -> [TVS1 shunt] -> D1 -> [C1 shunt] -> BUCK -> +5V OUT

Page 3 of 4 -- Split Design / Board B: Sensor Board

60 x 40 mm. 3.3V + I2C from Waveshare -> ADS1115 with sensor input protection. Stack-mounts above Board A on M3 standoffs.



BOARD B NOTES:

* J1 is a 4-pin JST PH (2.0 mm) matching the Waveshare's I2C JST.
  Pinout: 3V3 / GND / SDA / SCL.

* If using the bare ADS1115 chip instead of the breakout, wire ADDR pin to GND
  on this board (selects I2C address 0x48).

* TVS2 (BAT54S) clamps the SENSE node between GND and 3.3V -- protects the ADC's
  3.6V abs-max input from inductive spikes coupled onto the sender wire.

* R8 pulls SENSE up to 3.3V; the 240-30 ohm sender pulls it back down.
* R11 limits inrush if the sender ever shorts.

* Keep the analog sense path (R8 -> R11 -> J2) SHORT and away from any digital
  traces. Pour a GND copper region around it on the bottom side.

* Mounting holes at 3.5 mm from each corner on a 60 x 40 mm board place them at
  53 x 33 mm pitch. M3 standoffs let you stack this above Board A.

Page 4 of 4 -- BOM, Ordering, and Assembly Tips

Component sourcing for the hand-build single-board AND the EasyEDA two-board option.

BILL OF MATERIALS (combined; same parts for hand-build or split design)

Ref	Description	Notes
F1	Fuse, 1 A slow-blow ATO + PCB-mount holder	Littelfuse 0287000.PXCN or generic
D1	Diode, 1N5822 Schottky (40 V / 3 A)	axial DO-201
TVS1	TVS, 1.5KE24CA bidirectional	axial; load-dump clamp
C1	Cap, 220 uF / 35 V / 105 deg C electrolytic	radial, 8 mm dia
BUCK	DC-DC buck module, 12V to 5V, 1A min	LM2596 / MP1584 sub-module
ADS1115	ADC, ADS1115 16-bit I2C breakout	Adafruit / generic, 0x48 (ADDR=GND)
C2	Cap, 0.1 uF ceramic (X7R, 50 V)	ceramic disc or 0805 SMD
TVS2	TVS, BAT54S dual Schottky	SOT-23, low-leakage 3V3 clamp
R8	Resistor, 1 kohm 1% 1/4 W metal-film	pullup on SENSE node
R11	Resistor, 100 ohm 1% 1/4 W metal-film	series current limit
J1 (12V IN)	Screw terminal, 5.08 mm 2-position	Phoenix MPT or generic; rising-cage clamp
J2 (5V OUT)	Screw terminal, 3.5 mm 4-position	two paralleled rails (Wave + GPS)
J3 (I2C)	JST PH 4-pin connector (2.0 mm pitch)	matches Waveshare I2C JST
J4 (SENSOR)	Screw terminal, 5.08 mm 2-position	shielded sender wire
J5 (GPS)	JST 4-pin connector or 4-wire pigtail	+5V / GND / TX / RX
Stand-offs	M3 brass standoffs + screws (4x)	Power+Sensor stack on split design
Perfboard	125 x 80 mm prototype PCB	ELEGOO / Adafruit, 2.54 mm pitch

ORDERING TIPS

* DigiKey or Mouser is cheapest for resistors, caps, diodes, and connectors at any quantity. Amazon is fast for hobby-quantity ADS1115 breakouts and BUCK modules.

* For EasyEDA / JLCPCB fabrication, the 5-board minimum is ~$5 for a 100 mm board. Their PCB Assembly service can populate the small SMD parts (C2, TVS2 SOT-23) cheaply if you use their part library; leave bigger parts (BUCK, ADS1115, screw terminals) as through-hole for hand-soldering after the bare boards arrive.

* Buy ATO fuse holders that mount with two solder pads -- the PCB clip-style ones vibrate loose in cars over time.

* For screw terminals, get 5.08 mm pin pitch with 'rising cage' clamps (wire goes IN the side hole, not under a plate). They hold stranded automotive wire much better and don't cut individual strands when tightened.

* If you ever need to replace the BUCK with a 12V -> 3.3V regulator (skipping the Waveshare's USB path entirely), use a TPS54302 or LM2575 -- pin-compatible with most LM2596 modules but trims to 3.3V.